American Computer Science League

2025-2026 ● Contest 1: Short Problems Solutions ● Intermediate Division

1. Computer Number Systems

Evaluate in octal:

$2025_8 - 202_8 + 20_8 / 2_8 = 2025_8 - 202_8 + 10_8$
$= 1623_8 + 10_8$
$= 1633_8$

C. 1633

2. Computer Number Systems

The first 10 Fibonacci numbers when converted to binary are:

$0 \Rightarrow 0$	$1 \Rightarrow 1$	$1 \Rightarrow 1$
$2 \Rightarrow 10$	$3 \Rightarrow 11$	$5 \Rightarrow 101$
$8 \Rightarrow 1000$	$13 \Rightarrow 1101$	$21 \Rightarrow 10101$
$34 \Rightarrow 100010$		

Therefore there are 16 1s in the binary notation.

C. 16

3. Recursive Functions

A 16x16 square becomes an 8x8 square, then a 4x4, then a 2x2, then a 1x1, and finally a 0.5x0.5. Therefore, its perimeter is 4*0.5 = 2.

C. 2

4. Recursive Functions

$f(16, 12) = f(16 - 2, 12 + 1) + 12 = f(14, 13) + 12 = 7$
$f(14, 13) = f(14 - 2, 13 + 1) + 13 = f(12, 14) + 13 = -5$
$f(12, 14) = f(12 + 1, 14) - 12 = f(13, 14) - 12 = -6 - 12 = -18$
$f(13, 14) = f(13 + 1, 14) - 13 = f(14, 14) - 13 = 7 - 13 = -6$
$f(14, 14) = 14/2 = 7$ Now substitute backwards.
Also, $f(16, 12) = 7 - 13 - 12 + 13 + 12 = 7$

B. 7

5. What Does This Program Do?

```
input a, b, c
if c == 0 then
    c = sqrt(a^2 + b^2)
else
    if b == 0 then
        b = sqrt(c^2 - a^2)
    else
        a = sqrt(c^2 - b^2)
    end if
end if
output a,b,c
```

The program finds the missing value in a right triangle so in each case, the value that is 0 is replaced with the correct value. This problem is in honor of Pythagorean Theorem Day on 7/24/25.

D. All of the above

6. What Does This Program Do?

```
a = 13
b = 21
n = 8
while sqrt(b)!=int(sqrt(b))
    n = n + 1
    c = a + b
    a = b
    b = c
end while
output n
```

The loop will stop when 144 is a perfect square. It is the 12th Fibonacci number in the sequence 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144 since 21 is the 8th one.

A. 12